pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- RLS Policies
- Rules
- Triggers
- booking_flight
  - Columns
  - Constraints
  - Indexes
  - RLS Policies
  - Rules
  - Triggers
- flights
  - Columns (14)
    - flight_id integer
    - flight_no character varying(50)
    - scheduled_departure date
    - scheduled_arrival date
    - departure_airport_id integer
    - arrival_airport_id integer
    - departing_gate character varying(50)
    - arriving_gate character varying(50)
    - airline_id integer
    - status character varying(50)
    - actual_departure date
    - actual_arrival date
    - created_at date
    - update_at date
  - Constraints
  - Indexes
  - RLS Policies
  - Rules
  - Triggers
- passengers
  - Columns
  - Constraints
  - Indexes
  - RLS Policies
  - Rules
  - Triggers
- security_check

Dashboard | Properties | SQL | Statistics | Dependencies | Dependents | Processes | postgres/postgres... | airport_db.sql* | public.baggage/po...

postgres/postgres@PostgreSQL 18

No limit

Query | Query History

```
1  CREATE INDEX actual_departure_index ON flights (actual_departure);
```

Data Output | Messages | Notifications

```
CREATE INDEX

Query returned successfully in 45 msec.
```

Total rows: | Query complete 00:00:00.045 | CRLF | Ln 1, Col 67

18°C
Mostly cloudy

ENG 14:50 07.11.2026

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- update_at date
- Constraints
- Indexes
- RLS Policies
- Rules
- Triggers
- baggage
  - Columns (5)
    - baggage_id integer
    - weight_in_kg numeric(4,2)
    - created_date date
    - update_date date
    - booking_id integer
  - Constraints
  - Indexes
  - RLS Policies
  - Rules
  - Triggers
- baggage_check
  - Columns (6)
    - baggage_check_id integer
    - check_result character varying(50)
    - created_at date
    - update_at date
    - booking_id integer
    - passenger_id integer
  - Constraints
  - Indexes
  - RLS Policies
  - Rules
  - Triggers
- boarding_pass
- booking
  - Columns (7)
    - booking_id integer
    - passenger_id integer
    - booking_platform character varying(5
    - created_at date

Dashboard × Properties × SQL × Statistics × Dependencies × Dependents × Processes × postgres/postgres... × airport_db.sql* × public.baggage/po... ×

postgres/postgres@PostgreSQL 18

No limit

Query Query History

```
1  CREATE UNIQUE INDEX flight_no_act_dep_index ON flights (flight_no, actual_departure);
```

Data Output Messages Notifications

```
ERROR:  создать уникальный индекс "flight_no_act_dep_index" не удалось
Ключ (flight_no, actual_departure)=(BR-RS, 2023-11-02) дублируется.

ОШИБКА:  создать уникальный индекс "flight_no_act_dep_index" не удалось
SQL state: 23505
Detail: Ключ (flight_no, actual_departure)=(BR-RS, 2023-11-02) дублируется.
```

Total rows: 25 | Query complete 00:00:00.054 | CRLF | Ln 1, Col 86

19°C Partly sunny

ENG 15:08 07.11.2026

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- update_at date
- Constraints
- Indexes
- RLS Policies
- Rules
- Triggers
- baggage
  - Columns (5)
    - baggage_id integer
    - weight_in_kg numeric(4,2)
    - created_date date
    - update_date date
    - booking_id integer
  - Constraints
  - Indexes
  - RLS Policies
  - Rules
  - Triggers
- baggage_check
  - Columns (6)
    - baggage_check_id integer
    - check_result character varying(50)
    - created_at date
    - update_at date
    - booking_id integer
    - passenger_id integer
  - Constraints
  - Indexes
  - RLS Policies
  - Rules
  - Triggers
- boarding_pass
- booking
  - Columns (7)
    - booking_id integer
    - passenger_id integer
    - booking_platform character varying(5
    - created_at date

Dashboard × Properties × SQL × Statistics × Dependencies × Dependents × Processes × postgres/postgres... × airport_db.sql* × public.baggage/po... ×

postgres/postgres@PostgreSQL 18

No limit

Query Query History

```
1  CREATE INDEX combined_flights ON flights (departure_airport_id, arrival_airport_id);
```

Data Output Messages Notifications

```
CREATE INDEX

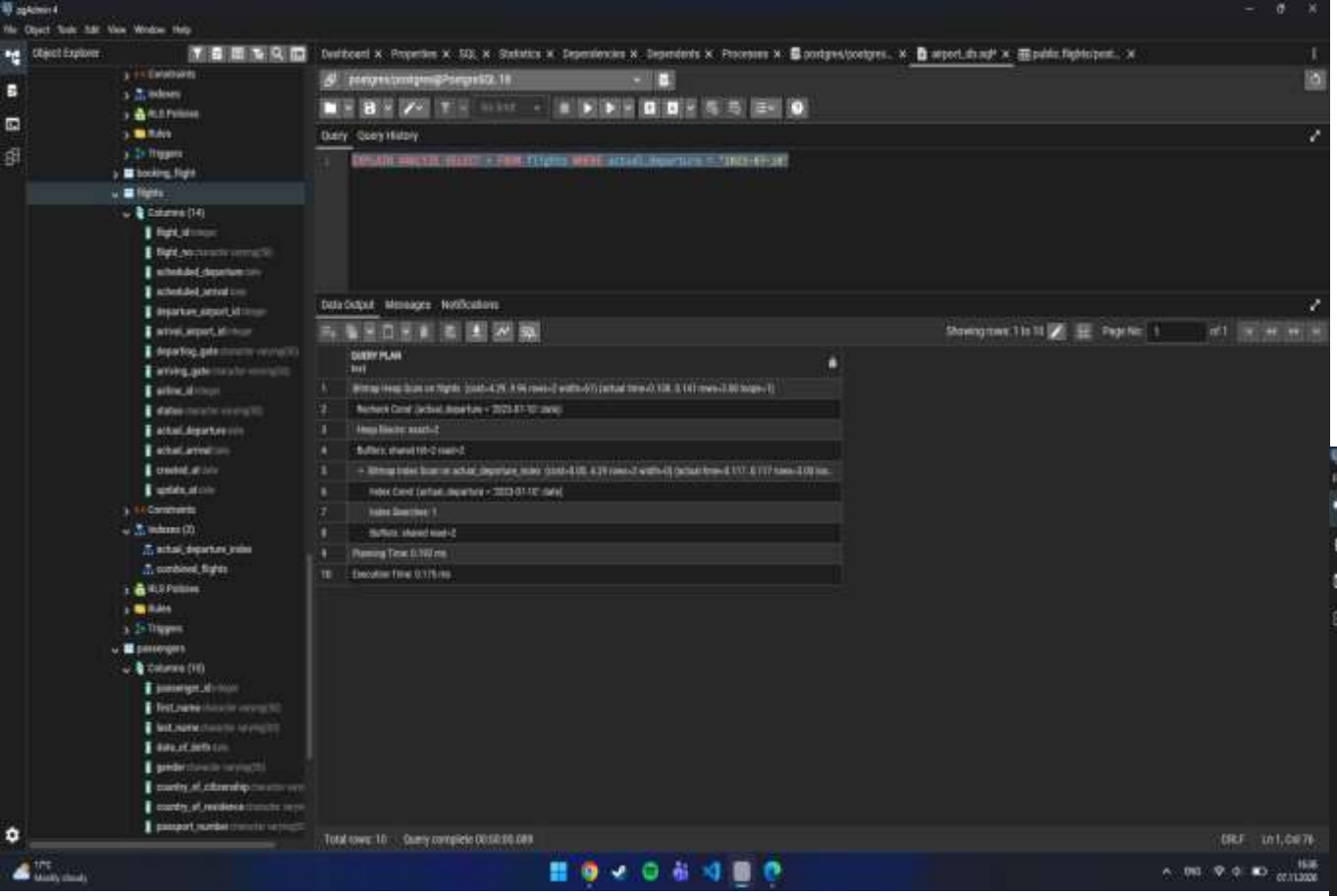
Query returned successfully in 50 msec.
```

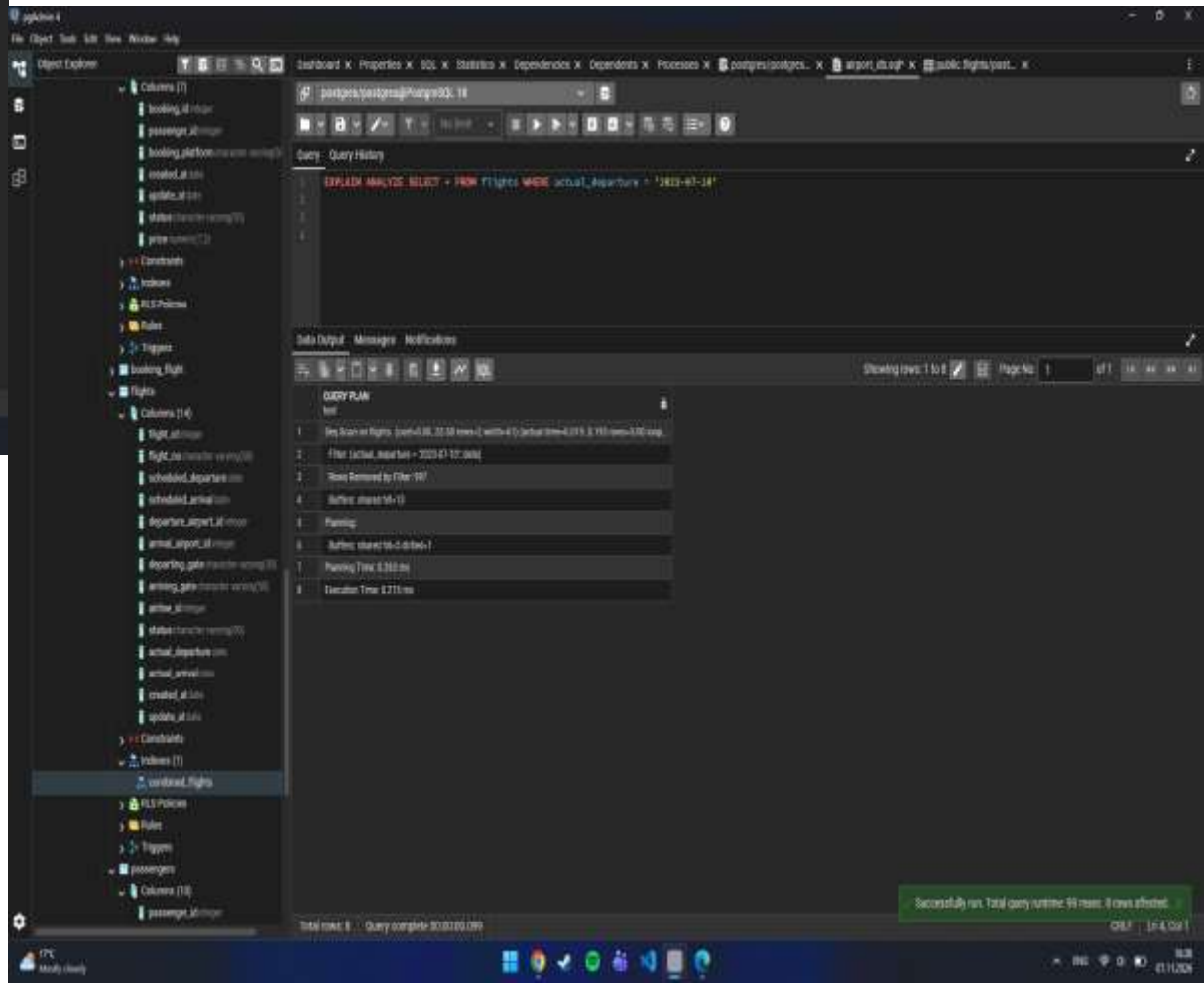
Query returned successfully in 50 msec.

Total rows: Query complete 00:00:00.050 CRLF Ln 1, Col 85

19°C Partly sunny

ENG 15:19 07.11.2026


pgAdmin 4
Object Explorer
Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
Processes
airport_db.sql*
public.flights/post...
Query
Query History
EXPLAIN ANALYZE SELECT * FROM flights WHERE actual_departure = '2023-07-10'
Data Output
Messages
Notifications
QUERY PLAN
text
Total rows: 10
Query complete 00:00:00.089
CRLF
Ln 1, Col 76


pgAdmin 4
Object Explorer
Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
Processes
airport_db.sql*
public.flights/post...
Query
Query History
EXPLAIN ANALYZE SELECT * FROM flights WHERE actual_departure = '2023-07-10'
Data Output
Messages
Notifications
QUERY PLAN
text
Successfully run. Total query runtime: 99 msec. 8 rows affected.

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Columns (7)
  - booking_id integer
  - passenger_id integer
  - booking_platform character varying(5
  - created_at date
  - update_at date
  - status character varying(50)
  - price numeric(7,2)
- Constraints
- Indexes
- RLS Policies
- Rules
- Triggers
- booking_flight
- flights
  - Columns (14)
    - flight_id integer
    - flight_no character varying(50)
    - scheduled_departure date
    - scheduled_arrival date
    - departure_airport_id integer
    - arrival_airport_id integer
    - departing_gate character varying(50)
    - arriving_gate character varying(50)
    - airline_id integer
    - status character varying(50)
    - actual_departure date
    - actual_arrival date
    - created_at date
    - update_at date
  - Constraints
  - Indexes (1)
    - combined_flights
  - RLS Policies
  - Rules
  - Triggers
- passengers
  - Columns (10)
    - passenger_id integer

Dashboard | Properties | SQL | Statistics | Dependencies | Dependents | Processes | postgres/postgres... | airport_db.sql* | public.flights/post...

postgres/postgres@PostgreSQL 18

No limit

Query | Query History

```
1  EXPLAIN ANALYZE SELECT * FROM flights
2  WHERE departure_airport_id  = 14 AND arrival_airport_id = 9
```

Data Output | Messages | Notifications

Showing rows: 1 to 10 | Page No: 1 of 1

	QUERY PLAN text
1	Bitmap Heap Scan on flights (cost=4.31..12.03 rows=3 width=61) (actual time=0.212..0.222 rows=4.00 loops=1)
2	Recheck Cond: ((departure_airport_id = 14) AND (arrival_airport_id = 9))
3	Heap Blocks: exact=4
4	Buffers: shared hit=4 read=2
5	-> Bitmap Index Scan on combined_flights (cost=0.00..4.31 rows=3 width=0) (actual time=0.177..0.178 rows=4.00 loo...
6	Index Cond: ((departure_airport_id = 14) AND (arrival_airport_id = 9))
7	Index Searches: 1
8	Buffers: shared read=2
9	Planning Time: 0.240 ms
10	Execution Time: 0.285 ms

Total rows: 10 | Query complete 00:00:00.113 | CRLF | Ln 2, Col 56

17°C Mostly cloudy

ENG 16:53 07.11.2026

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- arrival_airport_id integer
- departing_gate character varying(50)
- arriving_gate character varying(50)
- airline_id integer
- status character varying(50)
- actual_departure date
- actual_arrival date
- created_at date
- update_at date
- Constraints
- Indexes (1)
  - combined_flights
- RLS Policies
- Rules
- Triggers
- passengers
  - Columns (10)
    - passenger_id integer
    - first_name character varying(50)
    - last_name character varying(50)
    - date_of_birth date
    - gender character varying(50)
    - country_of_citizenship character varyi
    - country_of_residence character varyin
    - passport_number character varying(5(
    - created_at date
    - update_at date
  - Constraints
  - Indexes
  - RLS Policies
  - Rules
  - Triggers
- security_check
- Trigger Functions
- Types
- Views
- Subscriptions
- Login/Group Roles
- Tablespaces

Dashboard × Properties × SQL × Statistics × Dependencies × Dependents × Processes × postgres/postgres... × airport_db.sql* × public.passengers... ×

postgres/postgres@PostgreSQL 18

No limit

Query Query History

```
1
2  CREATE UNIQUE INDEX number_passp ON passengers (passport_number)
3  INSERT INTO passengers (passenger_id, first_name, last_name, date_of_birth, gender, country_of_citizenship, country_of_residence, passport_number, created_at, update_at
4  VALUES (201, 'Maksat', 'Erlanovich', '1995-02-07', 'Male', 'Kazakhstan', 'Kazakhstan', '11111111-8', '2018-10-29', '2019-09-15'),
5  (202, 'Max', 'Erlanovich', '1995-02-07', 'Male', 'Kazakhstan', 'Kazakhstan', '11111111-8', '2018-10-29', '2019-09-15');
```

Data Output Messages Notifications

```
ERROR:  повторяющееся значение ключа нарушает ограничение уникальности "number_passp"
Ключ "(passport_number)=(11111111-8)" уже существует.

ОШИБКА:  повторяющееся значение ключа нарушает ограничение уникальности "number_passp"
SQL state: 23505
Detail: Ключ "(passport_number)=(11111111-8)" уже существует.
```

Total rows: Query complete 00:00:00.062 CRLF Ln 1, Col 1

17°C Mostly cloudy

ENG 17:08 07.11.2026

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- arrival_airport_id integer
- departing_gate character varying(50)
- arriving_gate character varying(50)
- airline_id integer
- status character varying(50)
- actual_departure date
- actual_arrival date
- created_at date
- update_at date
- Constraints
- Indexes (1)
  - combined_flights
- RLS Policies
- Rules
- Triggers
- passengers
  - Columns (10)
    - passenger_id integer
    - first_name character varying(50)
    - last_name character varying(50)
    - date_of_birth date
    - gender character varying(50)
    - country_of_citizenship character varyi
    - country_of_residence character varyin
    - passport_number character varying(5(
    - created_at date
    - update_at date
  - Constraints
  - Indexes
  - RLS Policies
  - Rules
  - Triggers
- security_check
- Trigger Functions
- Types
- Views
- Subscriptions
- Login/Group Roles
- Tablespaces

Dashboard × Properties × SQL × Statistics × Dependencies × Dependents × Processes × postgres/postgres... × airport_db.sql* × public.passengers... ×

postgres/postgres@PostgreSQL 18

No limit

Query Query History

```
1  CREATE INDEX for_passengers ON passengers (first_name, last_name, date_of_birth, country_of_citizenship)
2  EXPLAIN ANALYZE
3  SELECT first_name || ' ' || last_name AS full_name FROM passengers WHERE country_of_citizenship = 'Philipines' AND date_of_birth >= '1984-01-01' AND date_of_birth < '19
```

Data Output Messages Notifications

SQL

Showing rows: 1 to 8 Page No: 1 of 1

	QUERY PLAN text
1	Seq Scan on passengers (cost=0.00..6.50 rows=1 width=32) (actual time=0.054..0.054 rows=0.00 loops=1)
2	Filter: ((date_of_birth >= '1984-01-01'::date) AND (date_of_birth < '1985-01-01'::date) AND ((country_of_citizenship)::text = 'Philipines':...
3	Rows Removed by Filter: 200
4	Buffers: shared hit=3
5	Planning:
6	Buffers: shared hit=6
7	Planning Time: 0.246 ms
8	Execution Time: 0.082 ms

Total rows: 8 Query complete 00:00:00.104 CRLF Ln 3, Col 179

17°C Mostly cloudy

ҚАЗ 17:22 07.11.2026

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- arrival_airport_id integer
- departing_gate character varying(50)
- arriving_gate character varying(50)
- airline_id integer
- status character varying(50)
- actual_departure date
- actual_arrival date
- created_at date
- update_at date
- Constraints
- Indexes (1)
  - combined_flights
- RLS Policies
- Rules
- Triggers
- passengers
- Columns (10)
  - passenger_id integer
  - first_name character varying(50)
  - last_name character varying(50)
  - date_of_birth date
  - gender character varying(50)
  - country_of_citizenship character varyi
  - country_of_residence character varyin
  - passport_number character varying(5(
  - created_at date
  - update_at date
- Constraints
- Indexes
- RLS Policies
- Rules
- Triggers
- security_check
- Trigger Functions
- Types
- Views
- Subscriptions
- Login/Group Roles
- Tablespaces

Dashboard × Properties × SQL × Statistics × Dependencies × Dependents × Processes × postgres/postgres... × airport_db.sql* × public.passengers... ×

postgres/postgres@PostgreSQL 18

No limit

Query Query History

```
1  SELECT indexname AS index_name, indexdef AS definition FROM pg_indexes WHERE tablename = 'passengers'
2  DROP INDEX number_passp,
3  for_passengers;
```

Data Output Messages Notifications

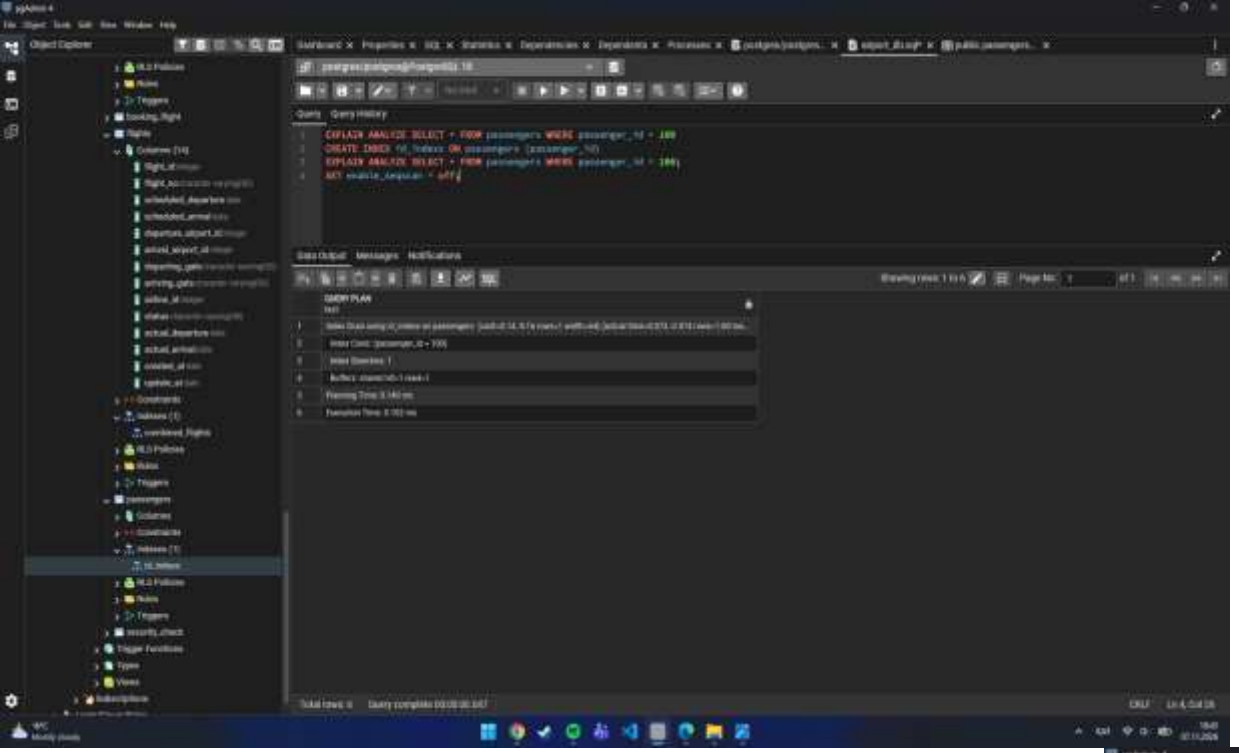
```
DROP INDEX

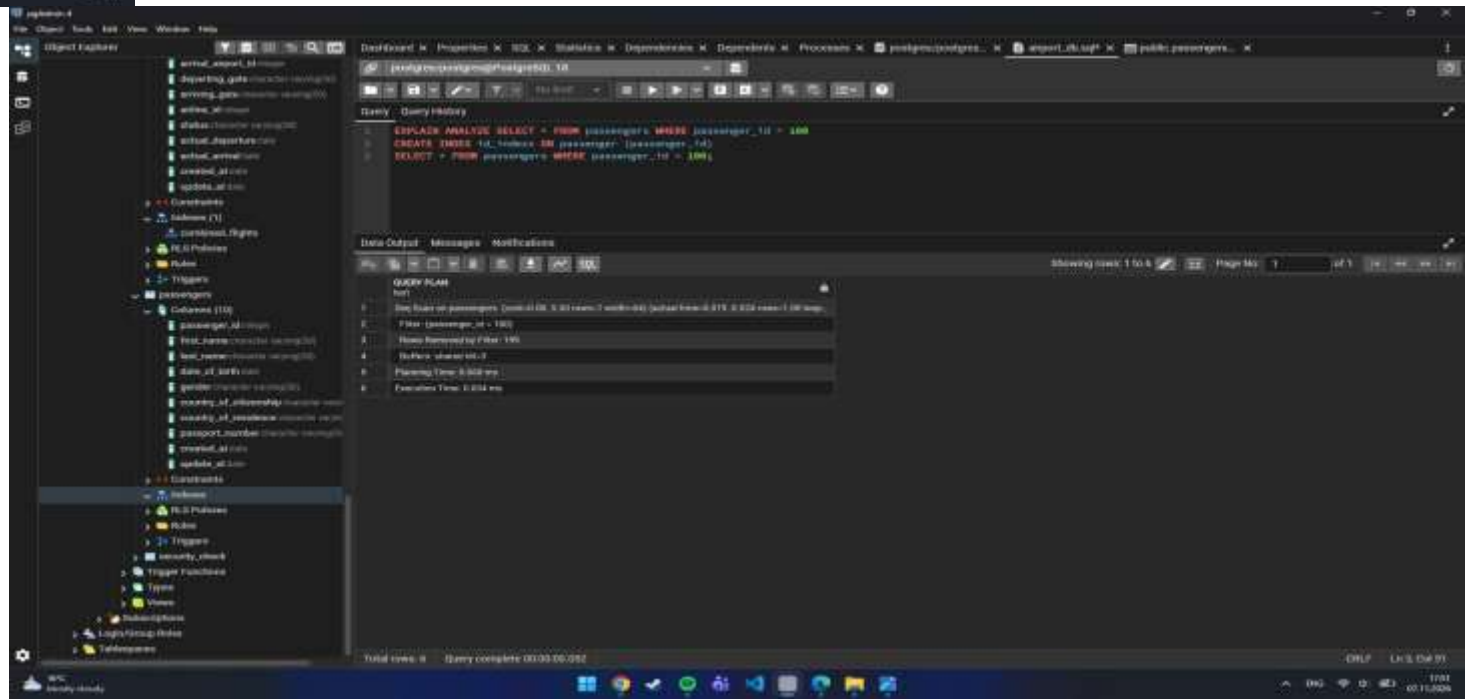
Query returned successfully in 53 msec.
```

Total rows: Query complete 00:00:00.053 CRLF Ln 3, Col 16

17°C Mostly cloudy

ENG 17:37 07.11.2026


EXPLAIN ANALYZE SELECT * FROM passengers WHERE passenger_id = 100
CREATE INDEX id_indexx ON passengers (passenger_id)
EXPLAIN ANALYZE SELECT * FROM passengers WHERE passenger_id = 100;
SET enable_seqscan = off;


EXPLAIN ANALYZE SELECT * FROM passengers WHERE passenger_id = 100
CREATE INDEX id_indexx ON passenger (passenger_id)
SELECT * FROM passengers WHERE passenger_id = 100;

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Triggers
- boarding_pass
- booking
  - Columns (7)
    - booking_id integer
    - passenger_id integer
    - booking_platform character varying(50)
    - created_at date
    - update_at date
    - status character varying(50)
    - price numeric(7,2)
  - Constraints
  - Indexes
  - RLS Policies
  - Rules
  - Triggers
- booking_flight
  - Columns (5)
    - booking_flight_id integer
    - booking_id integer
    - flight_id integer
    - created_at date
    - update_at date
  - Constraints
  - Indexes
  - RLS Policies
  - Rules
  - Triggers
- flights
  - Columns (14)
    - flight_id integer
    - flight_no character varying(50)
    - scheduled_departure date
    - scheduled_arrival date
    - departure_airport_id integer
    - arrival_airport_id integer
    - departing_gate character varying(50)
    - arriving_gate character varying(50)
    - airline_id integer

Dashboard | Properties | SQL | Statistics | Dependencies | Dependents | Processes | postgres/postgres... | airport_db.sql* | public.passengers...

postgres/postgres@PostgreSQL 18

No limit

Query | Query History

```
1  CREATE INDEX booking_index ON booking_flight (booking_id, flight_id)
2  EXPLAIN ANALYZE
3  SELECT f.flight_no, b.price
4  FROM booking_flight bf
5  JOIN booking b ON b.booking_id = bf.booking_id
6  JOIN flights f ON f.flight_id = bf.flight_id
7  WHERE b.price > 9000
```

Data Output | Messages | Notifications

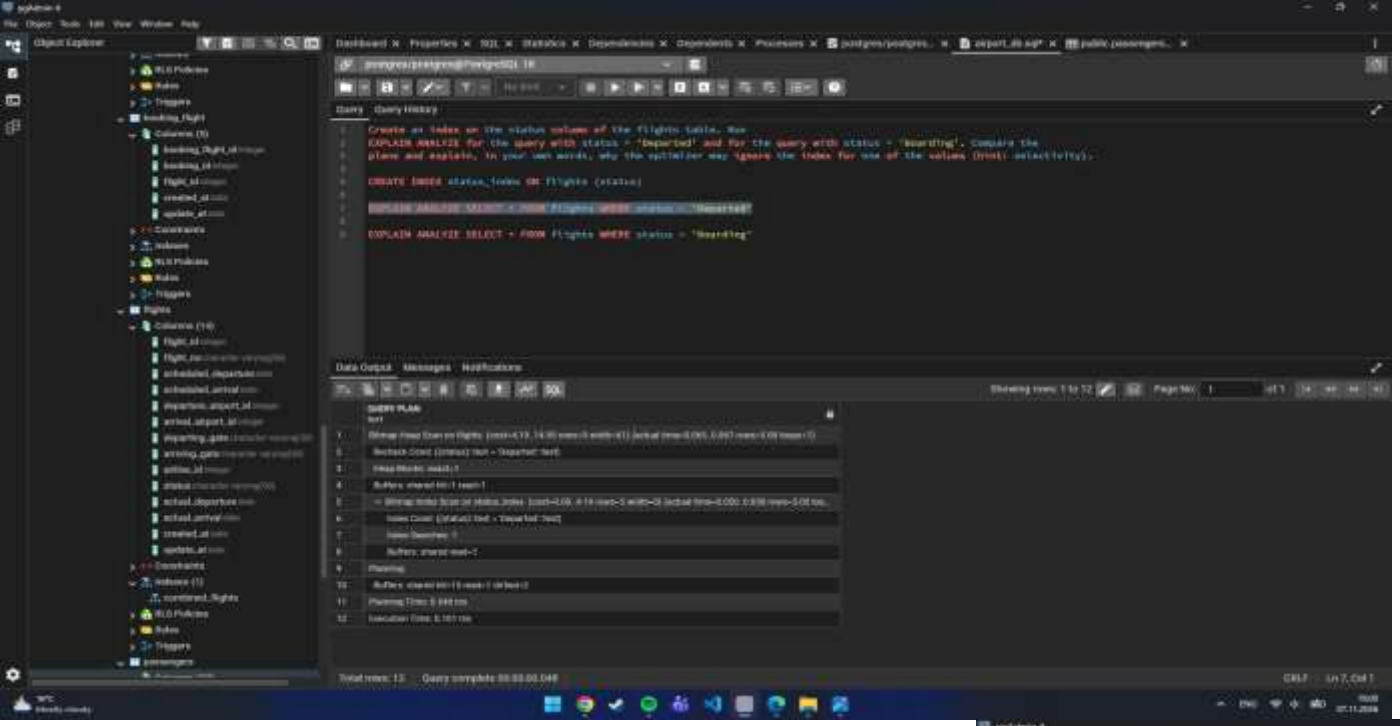
Showing rows: 1 to 24 | Page No: 1 of 1

	QUERY PLAN text
1	Hash Join (cost=33.15..61.10 rows=120 width=12) (actual time=0.448..0.643 rows=118.00 loops=1)
2	Hash Cond: (f.flight_id = bf.flight_id)
3	Buffers: shared hit=25
4	-> Seq Scan on flights f (cost=0.00..23.00 rows=1000 width=10) (actual time=0.020..0.097 rows=1000.00 loops=1)
5	Buffers: shared hit=13
6	-> Hash (cost=31.65..31.65 rows=120 width=10) (actual time=0.403..0.405 rows=118.00 loops=1)
7	Buckets: 1024 Batches: 1 Memory Usage: 13kB
8	Buffers: shared hit=12
9	-> Hash Join (cost=12.00..31.65 rows=120 width=10) (actual time=0.153..0.370 rows=118.00 loops=1)
10	Hash Cond: (bf.booking_id = b.booking_id)
11	Buffers: shared hit=12
12	-> Seq Scan on booking_flight bf (cost=0.00..17.00 rows=1000 width=8) (actual time=0.010..0.087 rows=1000.00 loo...
13	Buffers: shared hit=7
14	-> Hash (cost=11.25..11.25 rows=60 width=10) (actual time=0.121..0.122 rows=60.00 loops=1)
15	Buckets: 1024 Batches: 1 Memory Usage: 11kB
16	Buffers: shared hit=5
17	-> Seq Scan on booking b (cost=0.00..11.25 rows=60 width=10) (actual time=0.012..0.092 rows=60.00 loops=1)
18	Filter: (price > '9000'::numeric)
19	Rows Removed by Filter: 441
20	Buffers: shared hit=5
21	Planning:
22	Buffers: shared hit=18

Total rows: 24 | Query complete 00:00:00.055 | CRLF | Ln 7, Col 21

16°C Mostly cloudy

ENG 18:52 07.11.2026



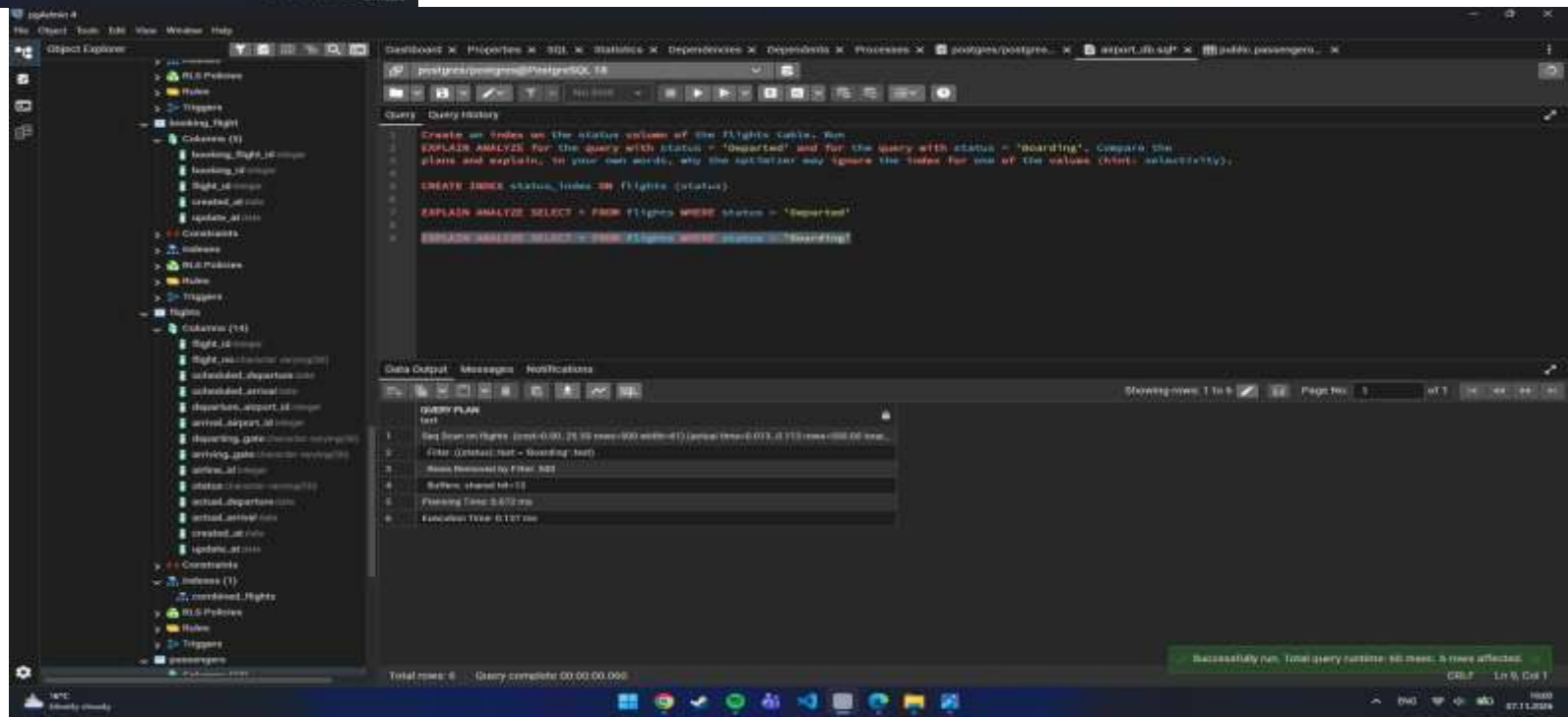
```
Create an index on the status column of the flights table. Run
EXPLAIN ANALYZE for the query with status = 'Departed' and for the query with status = 'Boarding'. Compare the
plans and explain, in your own words, why the optimizer may ignore the index for one of the values (hint: selectivity).

CREATE INDEX status_index ON flights (status)

EXPLAIN ANALYZE SELECT * FROM flights WHERE status = 'Departed'

EXPLAIN ANALYZE SELECT * FROM flights WHERE status = 'Boarding'
```

```
QUERY PLAN
text
Bitmap Heap Scan on flights  (cost=4.19..14.95 rows=5 width=47) (actual time=0.045..0.047 rows=0.00 loops=1)
  Recheck Cond: ((status)::text = 'Departed'::text)
  Heap Blocks: exact=1
  Buffers: shared hit=1 read=1
  ->  Bitmap Index Scan on status_index  (cost=0.00..4.19 rows=5 width=0) (actual time=0.030..0.030 rows=0.00 loo...
        Index Cond: ((status)::text = 'Departed'::text)
        Index Searches: 1
        Buffers: shared read=2
Planning
  Buffers: shared hit=15 read=1 dirtied=2
Planning Time: 0.848 ms
Execution Time: 0.101 ms
```



```
Create an index on the status column of the flights table. Run
EXPLAIN ANALYZE for the query with status = 'Departed' and for the query with status = 'Boarding'. Compare the
plans and explain, in your own words, why the optimizer may ignore the index for one of the values (hint: selectivity).

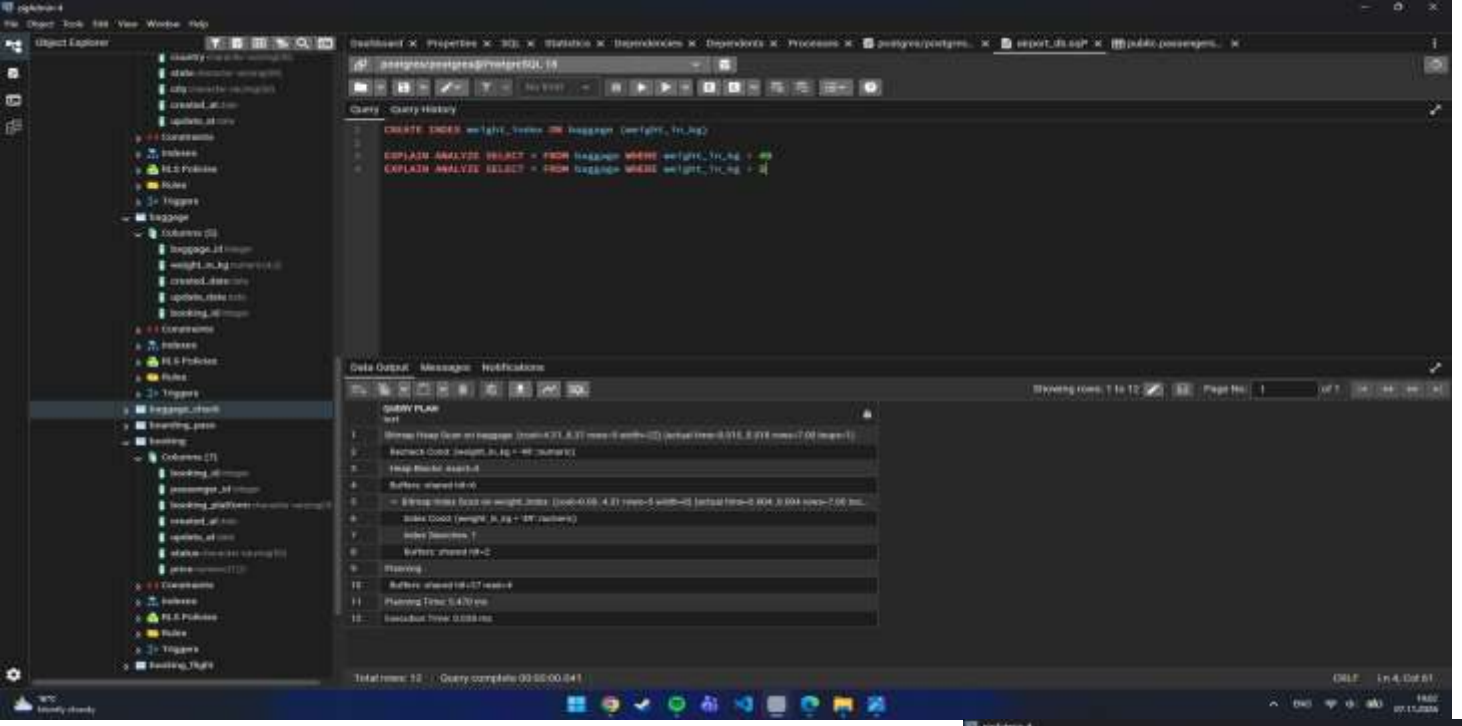
CREATE INDEX status_index ON flights (status)

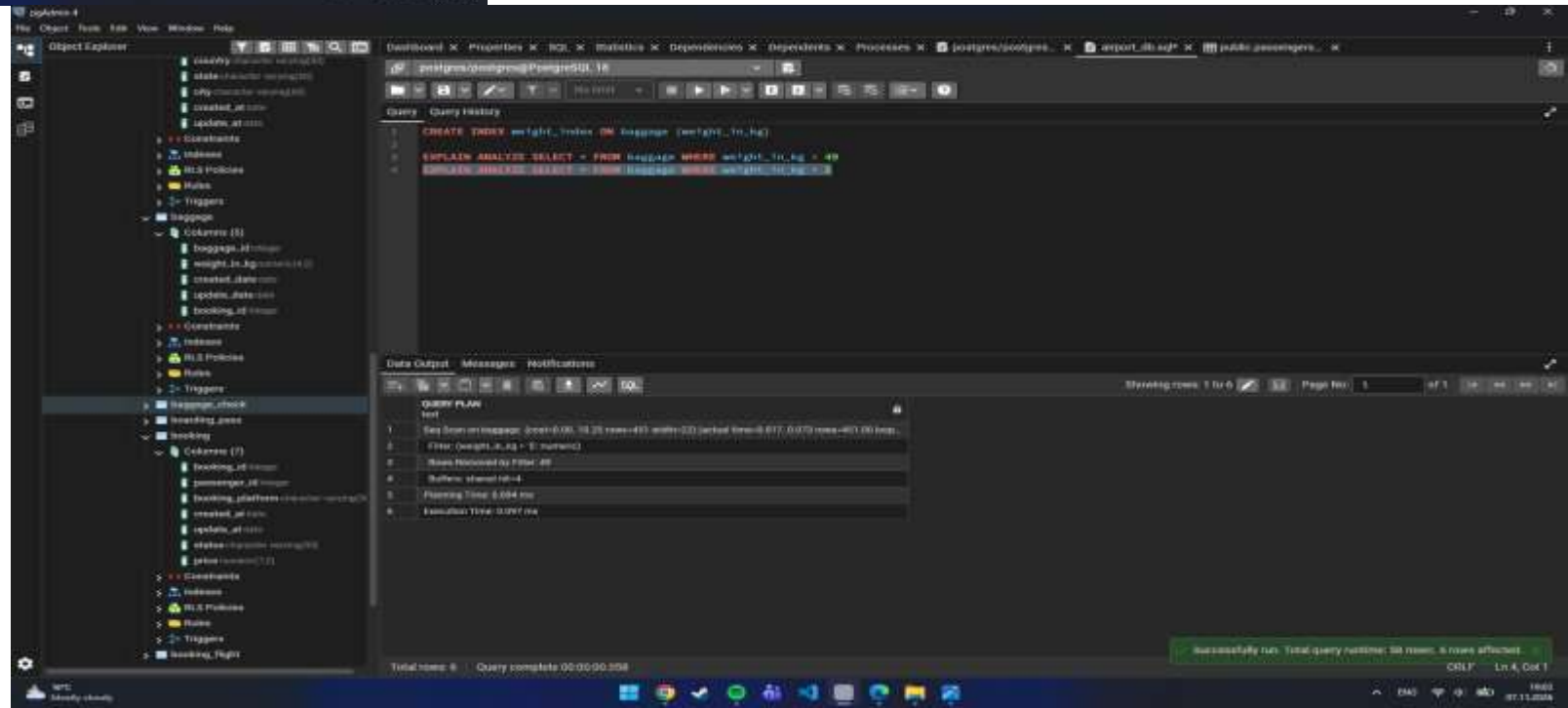
EXPLAIN ANALYZE SELECT * FROM flights WHERE status = 'Departed'

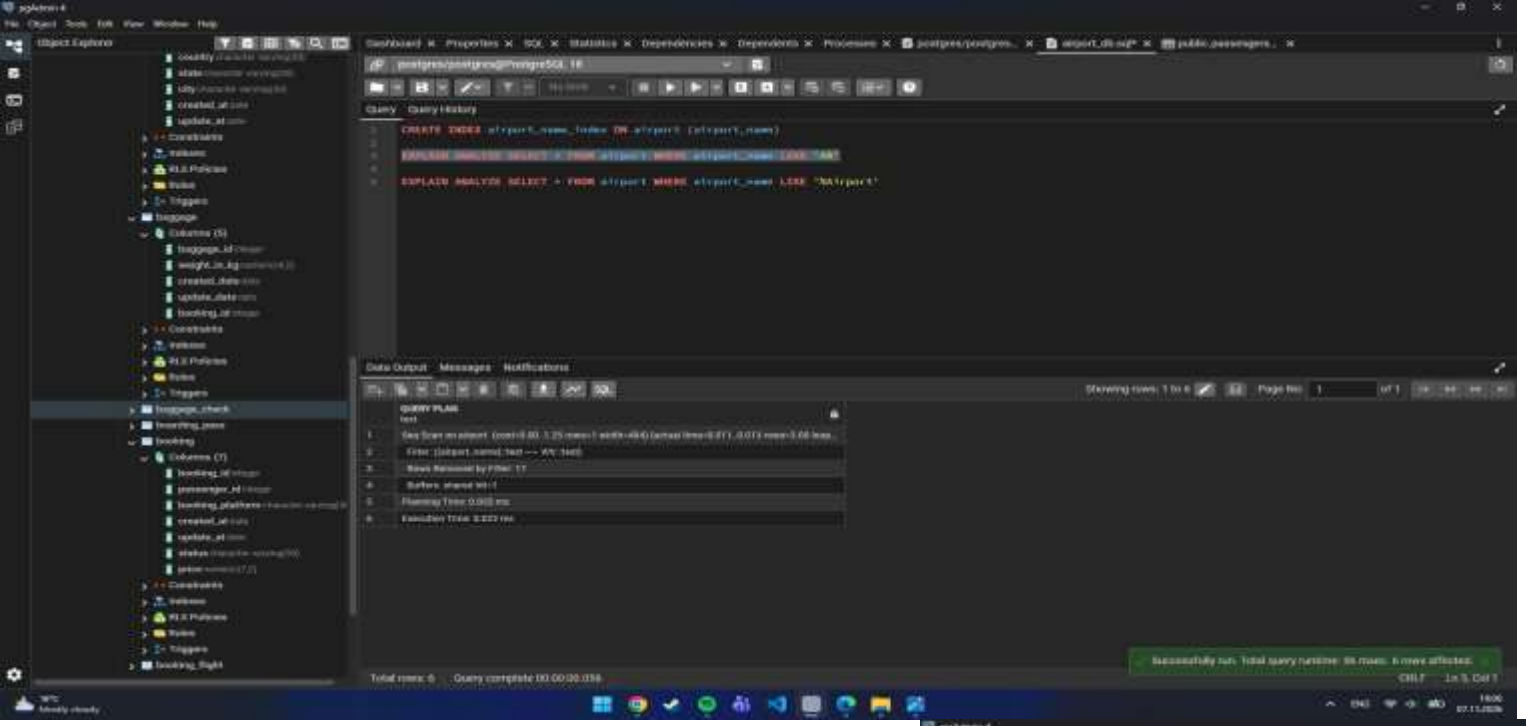
EXPLAIN ANALYZE SELECT * FROM flights WHERE status = 'Boarding'
```

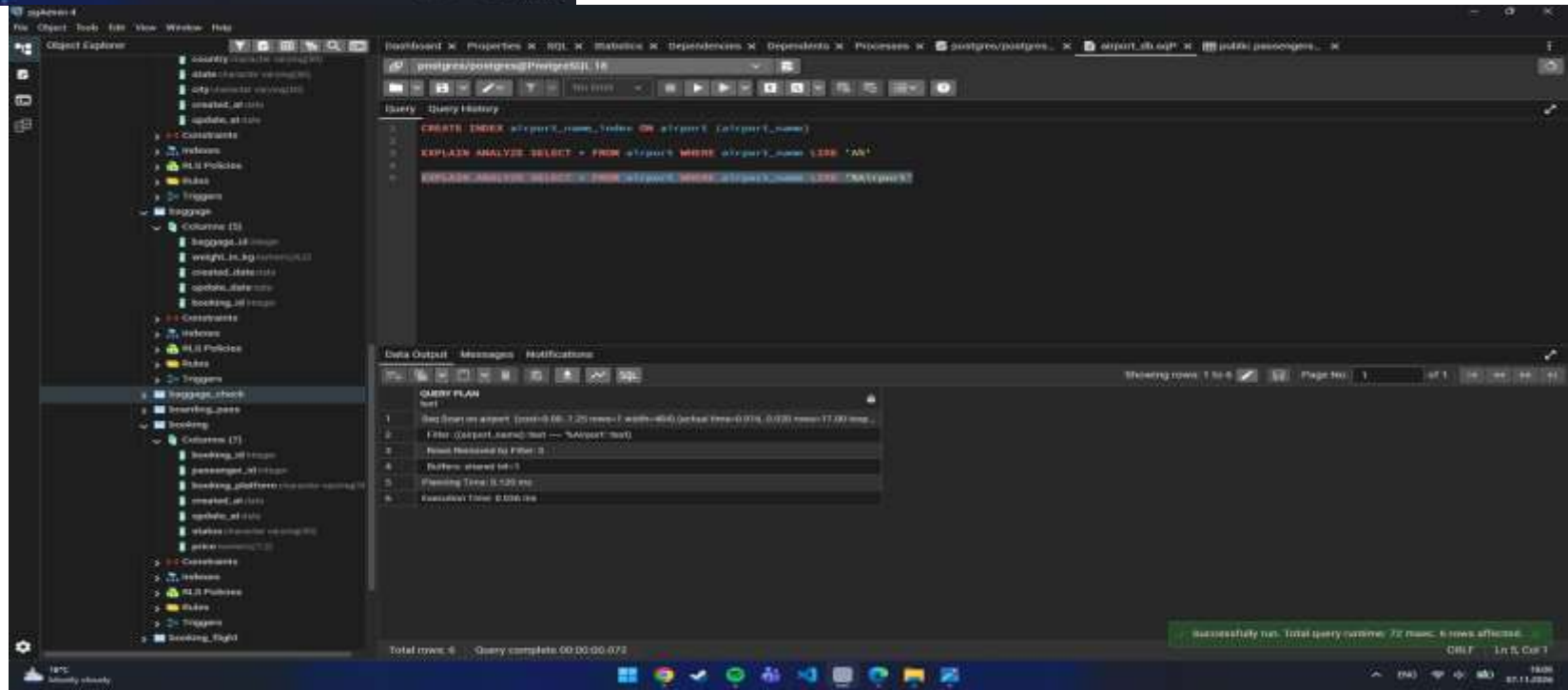
```
QUERY PLAN
text
Seq Scan on flights  (cost=0.00..29.50 rows=900 width=47) (actual time=0.013..0.113 rows=900.00 rows...
  Filter: ((status)::text = 'Boarding'::text)
  Rows Removed by Filter: 600
  Buffers: shared hit=15
Planning Time: 0.672 ms
Execution Time: 0.137 ms
```

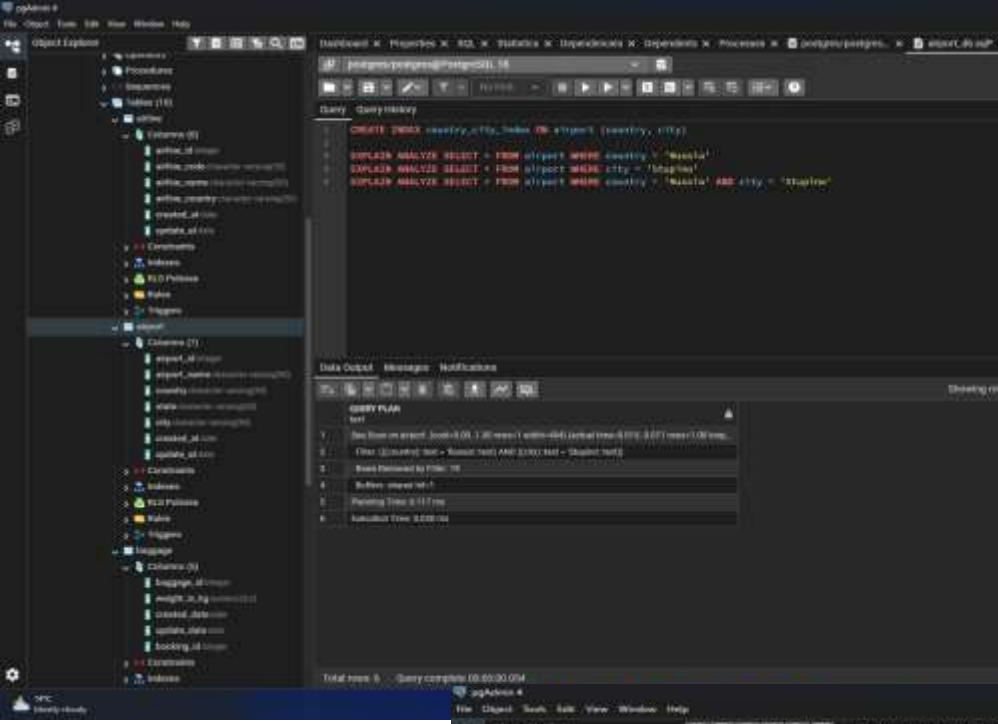
```
Successfully run. Total query runtime: 60 msec. 6 rows affected.
```


CREATE INDEX weight_index ON baggage (weight_in_kg)
EXPLAIN ANALYZE SELECT * FROM baggage WHERE weight_in_kg = 49
EXPLAIN ANALYZE SELECT * FROM baggage WHERE weight_in_kg = 3
Data Output
Messages
Notifications
QUERY PLAN
text


CREATE INDEX weight_index ON baggage (weight_in_kg)
EXPLAIN ANALYZE SELECT * FROM baggage WHERE weight_in_kg = 49
EXPLAIN ANALYZE SELECT * FROM baggage WHERE weight_in_kg = 3
Data Output
Messages
Notifications
QUERY PLAN
text

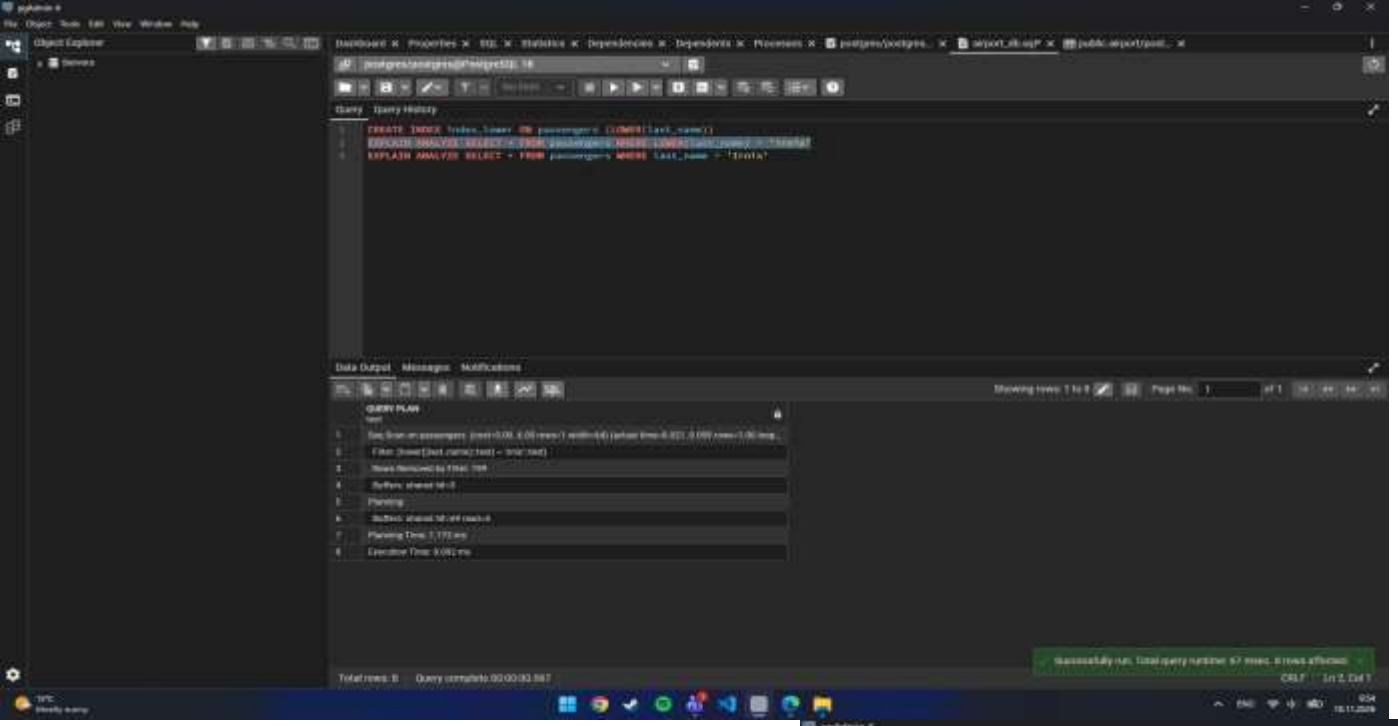

CREATE INDEX airport_name_index ON airport (airport_name)
EXPLAIN ANALYZE SELECT * FROM airport WHERE airport_name LIKE 'AN'
EXPLAIN ANALYZE SELECT * FROM airport WHERE airport_name LIKE 'NAirport'

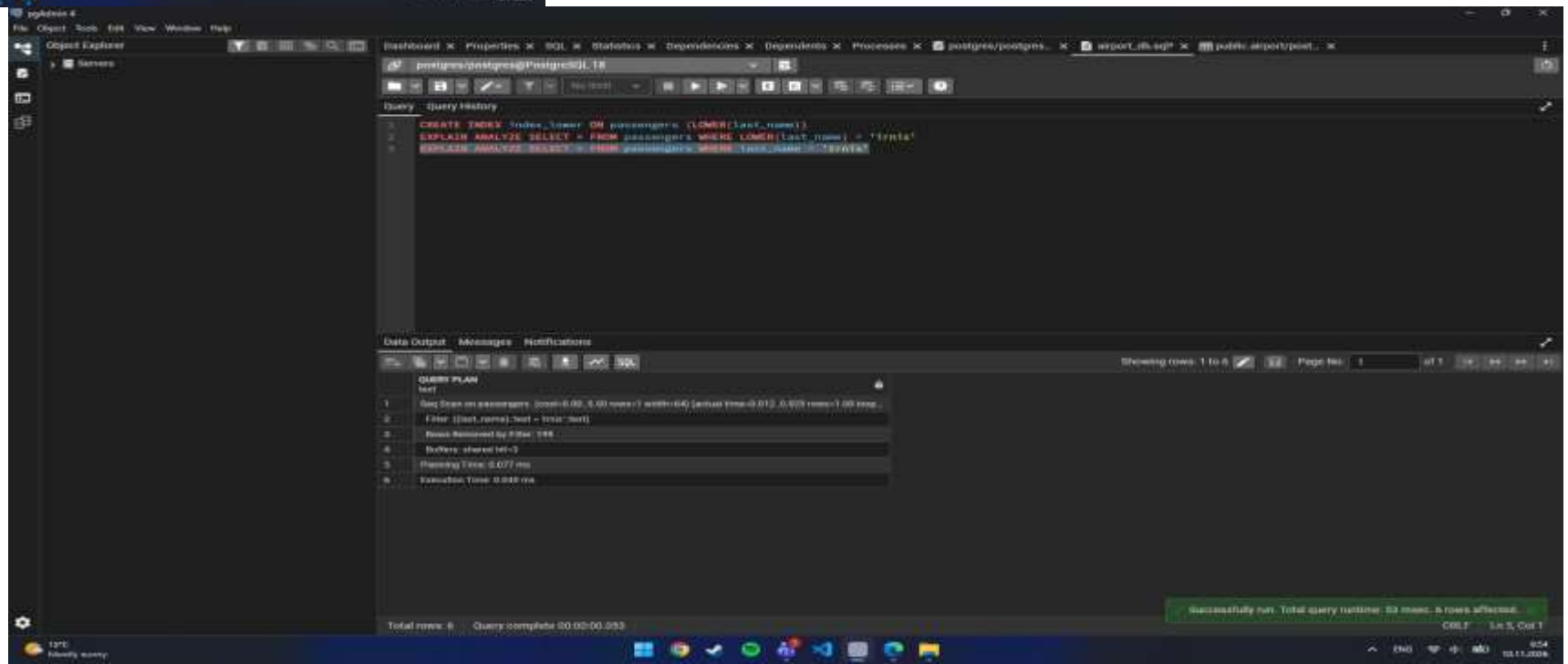

CREATE INDEX airport_name_index ON airport (airport_name)
EXPLAIN ANALYZE SELECT * FROM airport WHERE airport_name LIKE 'AN'
EXPLAIN ANALYZE SELECT * FROM airport WHERE airport_name LIKE 'NAirport'



```
CREATE INDEX country_city_index ON airport (country, city)

EXPLAIN ANALYZE SELECT * FROM airport WHERE country = 'Russia'
EXPLAIN ANALYZE SELECT * FROM airport WHERE city = 'Stupino'
EXPLAIN ANALYZE SELECT * FROM airport WHERE country = 'Russia' AND city = 'Stupino'
```


CREATE INDEX index_lower ON passengers (LOWER(last_name))
EXPLAIN ANALYZE SELECT * FROM passengers WHERE LOWER(last_name) = 'tirnta'
EXPLAIN ANALYZE SELECT * FROM passengers WHERE last_name = 'tirnta'
QUERY PLAN


CREATE INDEX index_lower ON passengers (LOWER(last_name))
EXPLAIN ANALYZE SELECT * FROM passengers WHERE LOWER(last_name) = 'tirnta'
EXPLAIN ANALYZE SELECT * FROM passengers WHERE last_name = 'tirnta'
QUERY PLAN

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

Servers

Dashboard Properties SQL Statistics Dependencies Dependents Processes postgres/postgres... airport_db.sql* public.airport/post...

postgres/postgres@PostgreSQL 18

No limit

Query Query History

```
1  CREATE INDEX index_passengers_price ON booking (passenger_id, price)
2  EXPLAIN ANALYZE SELECT price FROM booking WHERE passenger_id = 101
```

Data Output Messages Notifications

Showing rows: 1 to 12 Page No: 1 of 1

	QUERY PLAN text
1	Bitmap Heap Scan on booking (cost=4.30..9.36 rows=3 width=6) (actual time=0.052..0.055 rows=3.00 loops=1)
2	Recheck Cond: (passenger_id = 101)
3	Heap Blocks: exact=2
4	Buffers: shared hit=2 read=2
5	-> Bitmap Index Scan on index_passengers_price (cost=0.00..4.29 rows=3 width=0) (actual time=0.037..0.037 rows=3.00 loo...
6	Index Cond: (passenger_id = 101)
7	Index Searches: 1
8	Buffers: shared read=2
9	Planning:
10	Buffers: shared hit=38 read=2
11	Planning Time: 5.851 ms
12	Execution Time: 0.084 ms

Total rows: 12 Query complete 00:00:00.047 CRLF Ln 2, Col 1

15°C Mostly sunny

ENG 9:57 10.11.2026

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (10)
  - airline
    - Columns (6)
      - airline_id integer
      - airline_code character varying(50)
      - airline_name character varying(50)
      - airline_country character varying(50)
      - created_at date
      - update_at date
    - Constraints
    - Indexes
    - RLS Policies
    - Rules
    - Triggers
  - airport
    - Columns
    - Constraints
    - Indexes
    - RLS Policies
    - Rules
    - Triggers
  - baggage
  - baggage_check
  - boarding_pass
  - booking
  - booking_flight
  - flights
  - passengers
  - security_check
- Trigger Functions
- Types
- Views
- Subscriptions
- Login/Group Roles
- Tablespaces

Dashboard | Properties | SQL | Statistics | Dependencies | Dependents | Processes | postgres/postgres... | airport_db.sql* | public.airport/post...

postgres/postgres@PostgreSQL 18

No limit

Query | Query History

```
1  CREATE UNIQUE INDEX indexx ON airline (airline_code) WHERE airline_code IS NOT NULL
```

Data Output | Messages | Notifications

```
CREATE INDEX

Query returned successfully in 46 msec.
```

Total rows: | Query complete 00:00:00.046 | CRLF | Ln 1, Col 84

16°C Partly sunny

ENG 11:00 10.11.2026